

# Project 8: Uniform Omni-directional Magnetic Field Corrections using a Tri-Helical Coil

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## 1 Prospectus

Neutrinos are fundamental particles that are considered to be massless within the framework of the Standard Model. However, it has been shown from neutrino oscillation experiments that at least two of the three neutrino eigenstates must have nonzero mass [1]. Because what is measured is the squared mass difference between the neutrino generations, we have not been able to identify which of the three neutrino mass eigenstates  $m_1$ ,  $m_2$ , or  $m_3$  is the lightest, with these two possibilities being referred to as the normal ordering (NO) with  $m_1 < m_2 < m_3$  and inverted ordering (IO) with  $m_3 < m_1 < m_2$ . Current experiments put these differences at  $\Delta m_{21}^2 = (7.60 \pm 0.17) \times 10^{-5} \text{ eV}^2$  and  $\Delta m_{32}^2 = (2.445 \pm 0.023) \times 10^{-2} \text{ eV}^2$  (NO) or  $\Delta m_{32}^2 = (-2.52 \pm 0.04) \times 10^{-3} \text{ eV}^2$  (IO) [2].

Based on the results of KATRIN [3], it has been shown that the electron neutrino  $\nu_e$  has mass  $m_{\nu,e} < 450 \text{ meV}$ , which bounds the sum of neutrino masses as  $\sum m_\nu \lesssim 1.35 \text{ eV}$ . Their subsequent work aims to reduce the electron neutrino sensitivity threshold to below  $300 \text{ meV}$  [4]. Further bounds can be determined from cosmological model, with the current prevailing model,  $\Lambda\text{CDM}$ , giving  $\sum m_\nu < 130 \text{ meV}$  [5].

Project 8 seeks to measure the neutrino mass to much higher precision than what could be feasibly done by classical spectrometers such as KATRIN. Given that theory indicates  $m_{\nu,e} > 50 \text{ meV}$  in the case of IO, whereas  $m_{\nu,e} > 8 \text{ meV}$  is allowed for NO [2]. As such, Project 8 aims to measure the electron neutrino mass from beta decay by Cyclotron Radiation Emission Spectroscopy (CRES) to achieve a sensitivity of down to  $40 \text{ meV}$  [6], which would either let us find  $m_{\nu,e}$  or disprove IO.

The project requires a cold atomic tritium trap, which must be confined through the use of magnetic fields. In order to obtain such clear sensitivities, the magnetic field within the trap must be extremely uniform, with variations of less than 100 parts per billion. As such, I will model the magnetic field of the trap, accounting for correcting effects such as the sag due to gravity, to obtain a baseline for the system.

To show that these requirements are attainable, I plan to build upon the work of Gorton [7] by testing a configuration with a tri-helical coil, among other geometries. This may be accomplished by adding magnetic sensors to the inner region of the solenoid, modeling the current field within, calculating the necessary currents to pass through each coil, and programming the microprocessor to add those corrections. I would then verify that it accurately responds as intended.

## References

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